Lab Session 2

Mechanics applied to Aerospace Engineering
26/27-S1

Hugo Bergerioux

Plan for today

- **On report writing**
  - Main mistakes from first Lab session
  - Consistency

- **Problem of particle on oscillating loop**
  - Reminder of the problem
  - Main analytical solutions you need today

Report writing : Structure

- A report should **always start** with an **INTRODUCTION**
  - Introduce the problem to the reader
  - Present the **main objectives / challenges** of this work
  - Give the **plan** of the report (different sections etc…)

- It doesn't need to be long, keep it **concise and clear**, but it's mandatory

Globally good!

Report writing : Structure

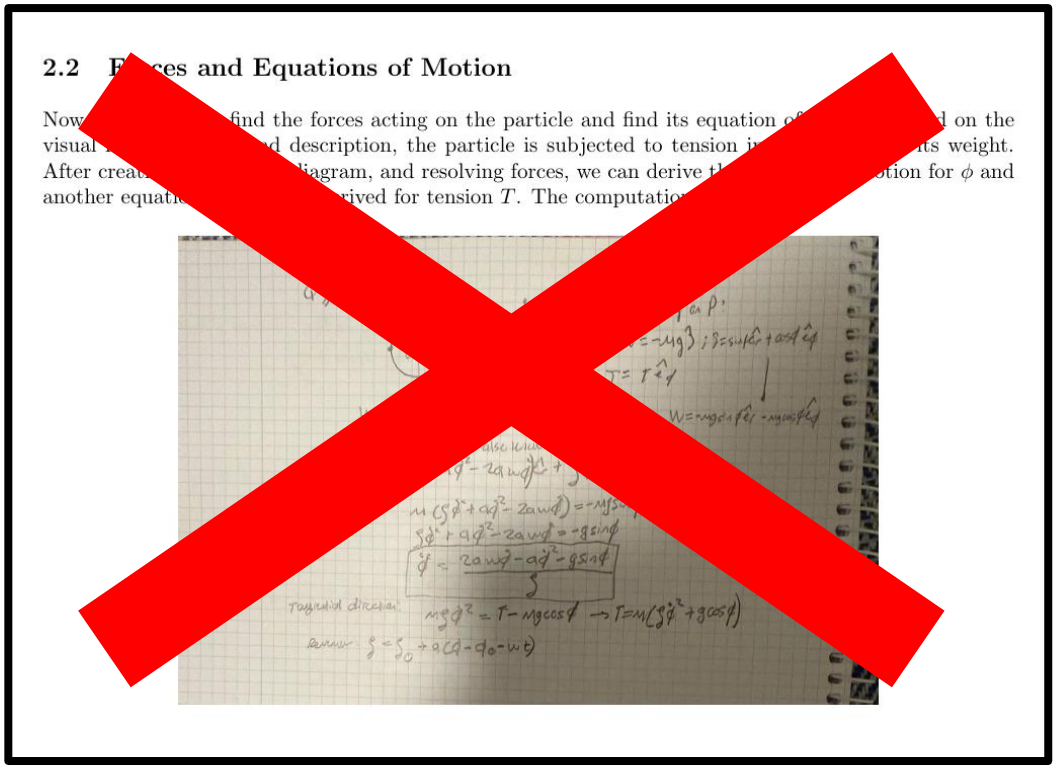
- The introduction should be followed by the **METHODOLOGY**
  - Present the methods you're using to solve the problem

- In this lab session, you should show :
  - The **complete analytical development**
    - Most of it has been done in class already
    - I will remind here the main equations needed today, but I expect the full development in the report
      - **This is evaluated as well**
  - The "numerical setup" : how you pass **from analytical to numerical implementation**

show the full derivation!

What not to do

- **Do not include screenshots** of your code / equations in the report, everything should be done numerically in LateX



2.2 F ces and Equations of Motion

Now find the forces acting on the particle and find its equation of d on the visual d description, the particle is subjected to tension its weight. After creat iagram, and resolving forces, we can derive th tion for ϕ and another equati rived for tension T. The computatio

With the generalised coordinate φ chosen, the kinematics follow. The total length of the string is the wrapped arc plus the free segment. The arc runs from the attachment point, which rotates with the spool, to B. When B moves counterclockwise the string unwinds, and when the spool turns counterclockwise it winds the string up. Using $\varphi(0) = 0$ and $\xi(0) = \xi_0$, the free length and its rate of change are

$$\xi = \xi_0 + a(\varphi - \omega t), \qquad \dot{\xi} = a(\dot{\varphi} - \omega). \tag{1}$$

Here and in the following, a dot denotes a time derivative: $\dot{\varphi} = \mathrm{d}\varphi/\mathrm{d}t$ is the angular rate at which the tangent point B moves around O, and $\ddot{\varphi} = \mathrm{d}^2\varphi/\mathrm{d}t^2$ is the corresponding angular acceleration.

The analysis is simpler in a basis that follows the point B. The unit vector $\mathbf{e}_r$ points from O to B, and $\mathbf{e}_\theta$ is obtained by rotating $\mathbf{e}_r$ ninety degrees counterclockwise:

$$\mathbf{e}_r = (\cos\varphi,\ \sin\varphi), \qquad \mathbf{e}_\theta = (-\sin\varphi,\ \cos\varphi). \tag{2}$$

The free segment of the string points from B to P, which is the direction of $-\mathbf{e}_\theta$. Both vectors turn with φ, so their time derivatives are

$$\dot{\mathbf{e}}_r = \dot{\varphi}\,\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\theta = -\dot{\varphi}\,\mathbf{e}_r. \tag{3}$$

How to present your results ?

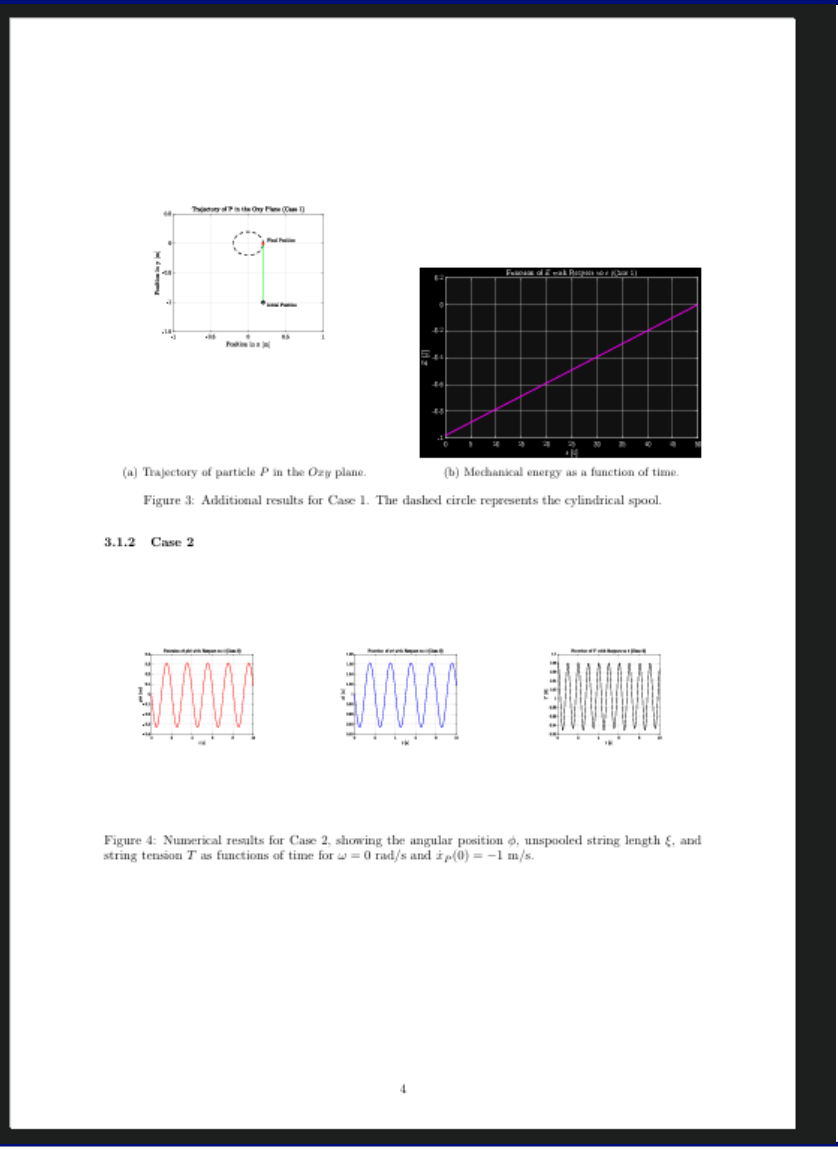
- Then you should present your **RESULTS** and **critically analyze / discuss** them in the report

- Your results (the figures) should be a tool (not a finality) to show what you want to emphasize
  - You're not just showing data, you're showing **physical phenomena**
    - So do not just put the plots there, but comment all of them
      - ***A plot without any comment is worthless***

What not to do

- **No screenshot of plots**, you can directly include the png/pdf versions within LateX

- The plots are usually the good size from Matlab run, but **then too small in the report**…
  - The figures need to be **readable**, use the space you have
  - **Avoid black background**



(a) Trajectory of particle P in the Oxy plane. (b) Mechanical energy as a function of time.

Figure 3: Additional results for Case 1. The dashed circle represents the cylindrical spool.

3.1.2 Case 2

Figure 4: Numerical results for Case 2, showing the angular position ϕ, unspooled string length ξ, and string tension T as functions of time for $\omega = 0$ rad/s and $\dot{x}_P(0) = -1$ m/s.

4

How to present your results ?

- The **DISCUSSION** part is very important : **you should comment the physics behind the data you show**
  - Please do not comment literally what you're seeing
  - No "the curve goes up"… everyone can see this, but I want you
    - To explain why it goes this way
    - To explain the consequence of this, what does it mean in our problem
- Please include the discussion **with the results**, after (or before) the figure
  - not at the very end
  - not in the caption of the figure
  - and reference the figure you're talking in the text
- In this report, **a lot of points go to the discussion**, so develop your argumentation and stay consistent !!

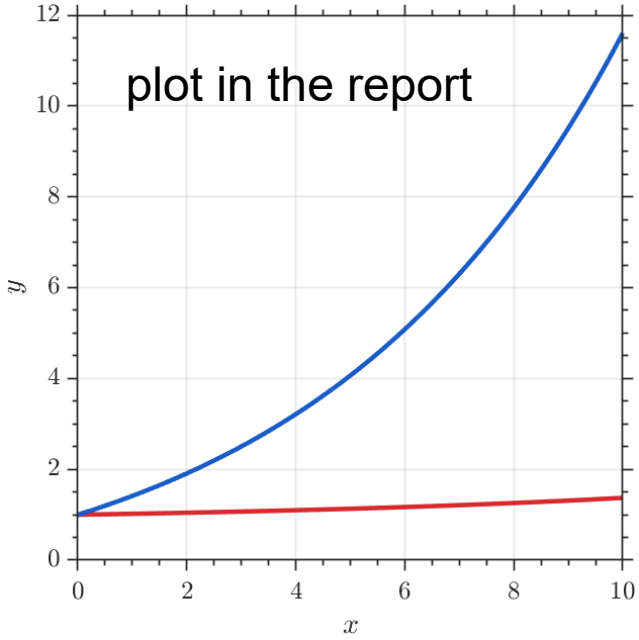
Consistency

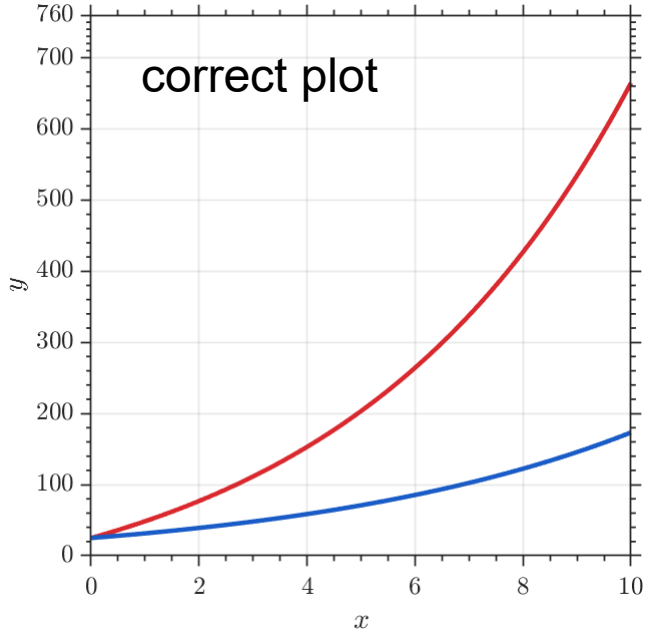
- It is important that your report **stays consistent**
  - usually clear copy/pasting of **AI is quite noticeable**

- 2 groups failed the first Lab because of the lack of consistency in their work
  - typically, a different equation presented in the report and implemented in the code…

Consistency

From last year, a lot of groups were contradicting in the discussion what they were showing in the figure





« as expected, the red curve increases faster than the blue one »

⇒ **NO !!**

Even if the results are wrong, stay consistent and acknowledge it

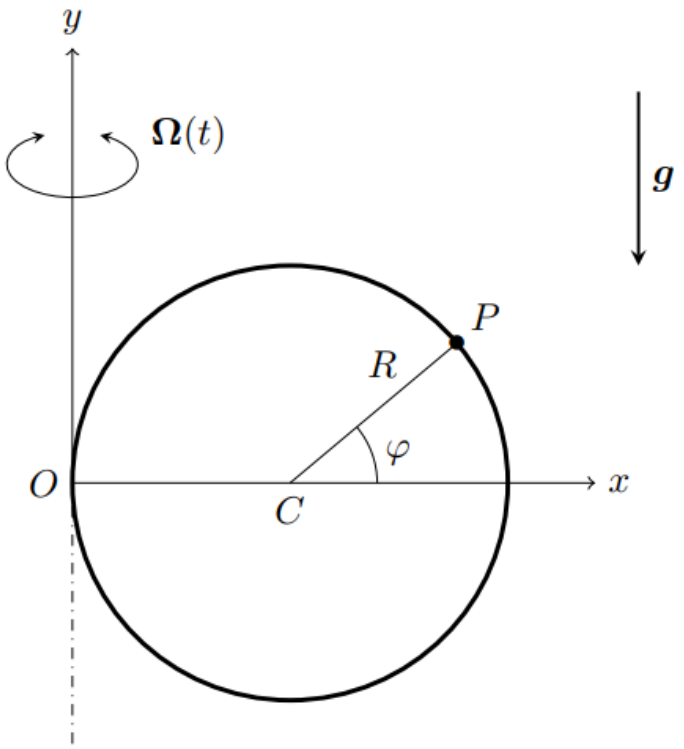
⇒ you'll lose less points

Report writing : General remarks

- Read carefully the Laboratory Guidelines
- Still use LaTeX for the report
  - **EXTRA POINT IF YOU DO IT WITH LATEX**
- You have **TWO WEEKS to submit** your results on Aula Global
  - Late submissions will be **penalized strongly**
  - Please follow submission instructions, it's important
    - Include your names in the report !
    - Indicate your group number as well

Problem : Particle on oscillating loop

A heavy point particle P of mass m slides without friction on a circular wire of center C and radius R. The wire is located on a vertical plane Oxy that rotates about the vertical axis Oy with a prescribed angular frequency $\boldsymbol{\Omega}(t) = A\omega \sin(\omega t)\boldsymbol{j}$, where A and ω are known positive constants, as shown in the figure. The angle that the vector $\boldsymbol{CP}$ makes with the Ox axis is denoted φ.



Problem : Analytical development

QUESTION 1, 2, 3 (already done in class)

- 2nd order differential equation

$$\ddot{\varphi} = -\frac{g}{R}\cos(\varphi) - A^2\omega^2 \sin^2(\omega t)[1+\cos(\varphi)]\sin(\varphi)$$

- Forces

$$N_1 = -mR\dot{\varphi}^2 + mg\sin(\varphi) - mA^2\omega^2 R\sin^2(\omega t)[1+\cos(\varphi)]\cos(\varphi)$$

$$N_2 = mA\omega R\left[2\dot{\varphi}\sin(\omega t)\sin(\varphi) - \omega\cos(\omega t)(1+\cos(\varphi))\right]$$

$$\boldsymbol{F}_{\text{Cor}} = -2mA\omega R\dot{\varphi}\sin(\omega t)\sin(\varphi)\mathbf{e}_z$$

$$\boldsymbol{F}_{\text{Euler}} = mA\omega^2 R\cos(\omega t)[1+\cos(\varphi)]\mathbf{e}_z$$

$$\boldsymbol{F}_{\text{cent}} = mA^2\omega^2 R\sin^2(\omega t)[1+\cos(\varphi)][\cos(\varphi)\mathbf{e}_r - \sin(\varphi)\mathbf{e}_\varphi]$$

Problem : Numerical implementation

QUESTION 4

- Similar implementation as what you did for Lab 1 on MATLAB
  - Define state vector

$$X = \begin{bmatrix} \varphi \\ \dot{\varphi} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

  - We need to solve 2 ODEs simultaneously :
    - $\dot{X}_1 = X_2$
    - $\dot{X}_2 = f(t, X_1, X_2)$

QUESTION 5

- Velocity term for the kinetic energy : $\boldsymbol{v} = \dot{\boldsymbol{r}} + \boldsymbol{\Omega} \times \boldsymbol{r}$

Problem : Trajectory and stability

QUESTION 6

- Plot trajectory of the particle in a stationary reference frame
  - *You can include the video of the animation of the trajectory in the .zip file you submit (**bonus points**)*

QUESTION 7

- Consider no oscillation of the loop
  - Study **impact of initial conditions on stability** of the solution
  - **Derive analytically the limit criteria** (follow the hint, it will help)
    - I expect then a validation of the analytical criteria obtained with the plots you obtained for the given initial conditions
    - Last year only a few groups did this

Problem : Linearization

QUESTION 8

- Linearized differential equation

$$\ddot{\delta} + \delta \frac{g}{R} - \Omega^2 = 0$$

- Highlight in the results the difference between full and linearized differential equation
  - Remember the physical phenomenon that you saw in class to explain the observed behavior

In general, I expect critical discussion of all the results you present !

Thank you!

Contact:

hbergeri@ing.uc3m.es

Please **sign the attendance sheet**
and keep the same team as for Lab 1

